GRE Quantitative Reasoning

모의고사 Set 15 — 문제지

최신 문제(뒷번호) · S2 고난도

구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내
- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 15 해설지)에 있습니다.

답안 기록표

S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
답															

Section 1 12 Questions · 21 Minutes

Question 1 대소비교

1273. Quantity A: The area of a rectangle with a perimeter of 32
Quantity B: The area of a rectangle with a perimeter of 52

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

441. Quantity A: The least integer x that satisfies the inequality $x^3 > -30$
Quantity B: -3

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

913. k is an integer.
Quantity A: The remainder when k^2-k is divided by 2
Quantity B: 0

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

918. The integer N is greater than 1,000.
Quantity A: The remainder when N is divided by 3
Quantity B: The remainder when N is divided by 17

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 객관식

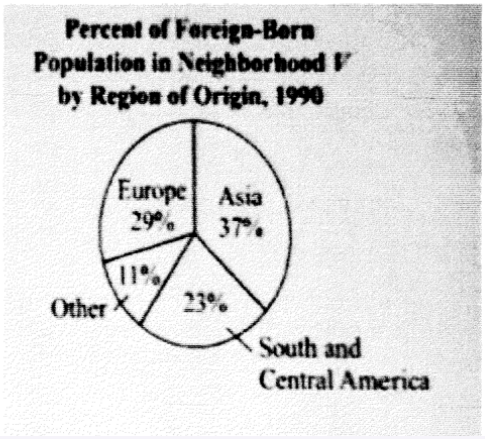
1001. Four slips of paper are numbered 1, 2, 3, and 4, respectively and put into an empty box. If two of these slips of paper will be drawn out of the box at random and without replacement, what is the probability that the sum of the two numbers written on the slips of paper will be 3?
A1/12
B1/6
C1/5
D1/3
E1

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Questions 6 to 8 are based on the following data.

Population Data for Five Neighborhoods of City X, 1930 and 1990

Neighborhood	Total Population		Foreign-Born Population	
	1930	1990	1930	1990
R	228,100	217,400	91,200	66,600
S	219,100	100,700	78,700	11,700
T	102,800	150,100	20,200	25,400
U	79,700	47,300	43,100	18,200
V	5,600	9,100	1,700	4,900
Total	635,300	524,600	234,900	126,800



Question 6 객관식

543. In 1990 in neighborhood V, 1/3 of the foreign-born population whose region of origin was Asia came from China. Approximately what fraction of the entire foreign-born population in neighborhood V came from China?
A1/12
B1/8
C1/5
D1/4
E1/3

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 7 객관식

544. A list of the names of the people of the entire 1990 foreign-born population in neighborhood V was generated, with each person's name appearing once.The names of 2 different people will be randomly selected from the list. Which of the following is closest to the probability that both names selected will be names of people whose region of origin was "Other"?
A0.01
B0.11
C0.25
D0.39
E0.49

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 8 객관식

545. In 1990 approximately what percent of the total population of neighborhood V was foreign-born with Europe as region of origin?
A6%
B12%
C16%
D25%
E29%

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 9 객관식

737. Solution A contains x percent alcohol by volume. Solution B contains y percent alcohol by volume. If 3 liters of solution A are mixed with 5 liters of solution B to yield 8 liters of solution C, what percent of the volume of solution C is alcohol?
A(3x+5y)/8
B(3/x+y/5)/8
C(3/x+5/y)/8
D8(3x+5y)/100
E8(3x+5y)

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 10 주관식

1081. If x and y are positive integers and $\{(8)(7)(6)(5)(4)(3)\}/\{(2^x)(3^y)\}$is an integer, what is the greatest possible value of xy?

Question 11 복수선택

1156. Which of the following could be a factor of $9!/\{(6!)(3!)\}$?
Indicate all such numbers.
A.2
B.3
C.4
D.5
E.6
F.7
G.8
H.9

Indicate all that apply. A B C D E F G H

Question 12 객관식

868. Light travels at approximately 186,000 miles per second. Which of the following is closest to the number of miles that light travels in 3 hours?
A200,000
B2,000,000
C20,000,000
D200,000,000
E2,000,000,000

A B C D E

Section 2 15 Questions · 26 Minutes

Question 1 대소비교

437. Quantity A: The perimeter of a rectangular region that has an area of 20
Quantity B: 20

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

1145. The function f is defined by $f(n)=3n^2+2n+5$ for all integers n.
Quantity A: The least value of f(n) for all integers n
Quantity B: 5

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

~~337. 문제 중복 삭제~~

338. n is an integer and 5n−1 is a positive even integer
Quantity A: $(-1)^{n+1}$
Quantity B: 1

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

1224. A 50-question survey was administered to 1,000 college students. The answer to each question was either yes or no. Each student was given a score on the survey equal to the number of questions to which the student answered yes.
Quantity A: The 90th percentile of the 1,000 scores
Quantity B: 45

(A) Quantity A is greater.

(B) Quantity B is greater.

(C) The two quantities are equal.

(D) The relationship cannot be determined from the information given.

Question 5 대소비교

1104. The median value for a sample of 7 numbers is 50.
The median value for a sample of 9 numbers is 52.
Quantity A: The median value for the combined sample of the 16 numbers
Quantity B: 51

(A) Quantity A is greater.

(B) Quantity B is greater.

(C) The two quantities are equal.

(D) The relationship cannot be determined from the information given.

Question 6 객관식

411. When positive integer m is divided by 6, the remainder is 4. When positive integer p is divided by 6, the remainder is 5. What is the remainder when the product mp is divided by 6?
A1
B2
C3
D4
E5

(A) (B) (C) (D) (E)

Question 7 객관식

430. The sum of two numbers divided by 2, gives a result of 24 and their difference divided by 2 gives a result of 17. The product of these two numbers is divisible by which of the following?
A4
B6
C34
D41
E51

(A) (B) (C) (D) (E)

Question 8 객관식

1210.

SOURCES OF FINANCIAL SUPPORT FRO A NONPROFIT ORGANIZATION

Source	Percent of Financial Support
Corporations	37%
Federal/state governments	33%
Individuals	22%
Others	8%

The table shows the percent distribution of financial support for a nonprofit organization, by source. The amount of financial support from individuals is approximately what percent less than the amount from corporations?
A.24%
B.35%
C.41%
D.69%
E.88%

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 9 복수선택

591. Working continuously for a total of 120 hours, a machine first assembled x units of product A and then assembled y units of product B. where x and y are positive integers. It took the machine 3 hours to assemble each unit of product A and 5 hours to assemble each unit of product B. Which of the following could be the total number of units of product A and product B that the machine assembled in the 120 hours?
Indicate all such numbers.
A22
B33
C38

Indicate all that apply. [A] [B] [C]

Question 10 주관식

1126. The width and the length of a rectangular piece of plywood are 4 feet and 8 feet, respectively. Along one edge of the plywood, a strip x inches wide and 8 feet long is removed. Then, along an edge perpendicular to the 8-foot edge, a strip x inches wide is removed. For what value of x will the remaining rectangular piece have width and length in the ratio of 2 to 5? (1 foot = 12 inches)

Question 11 객관식

499. If $-x < y < 0$, which of the following statements is true about the three quantities $(x+y)^2$, $y^2 - x^2$, and $x^2 - y^2$?
A$x^2 - y^2 < y^2 - x^2 < (x+y)^2$
B$x^2 - y^2 < (x+y)^2 < y^2 - x^2$
C$(x+y)^2 < y^2 - x^2 < x^2 - y^2$
D$y^2 - x^2 < x^2 - y^2 < (x+y)^2$
E$y^2 - x^2 < (x+y)^2 < x^2 - y^2$

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 12 객관식

716. A certain charity is conducting a fund-raiser. For the first $9,000 raised by the charity, Company B will contribute $1 for every $3 collected by the charity. For any amount over $9,000 raised by the charity, Company B will contribute $2 for every $5 collected by the charity. How much money must the charity raise in order to reach a total of $68,000, including the contribution from Company B?
A$34,000
B$40,000
C$45,000
D$49,000
E$56,000

A B C D E

Question 13 복수선택

755. Vladimir invested $10,000 for one year. He invested some of the amount at 4 percent simple annual interest and the rest of the amount at 6 percent simple annual interest.
If the total interest earned for the year was between $450 and $550, which of the following statements must be true?. Indicate all such statements.
AThe amount invested at 6 percent simple annual interest was greater than $2,000.
BThe amount invested at 6 percent simple annual interest was less than $8,000.
CThe amount invested at 6 percent simple annual interest was more than 3 times the amount invested at 4 percent simple annual interest.

Indicate all that apply. A B C

Question 14 객관식

1008. The integer n is the product of four different prime numbers. If n divided by 35 is a multiple of 13, which of the following could be equal to n divided by 7?
A49
B91
C130
D245
E260

A B C D E

Question 15 객관식

1129. The ratio of the number of males to the number of females in a mathematics class is 4 to 5. The corresponding ratio in an English class is 5 to 4. There are 50 percent more males in the English class than there are in the mathematics class. What is the ratio of the number of females in the mathematics class to the number of females in the English class?
A5:4
B4:5
C24:25
D25:24
E1:1

A B C D E